

WEEK 7 – Power BI

Student Performance – Data Cleaning & Transformation

- **APARNA F**
B.Sc Data Science & AI

Introduction

Week 7 focused on **data cleaning, transformation, and preparation of the Student Performance dataset using Power BI**. The main objective was to convert the raw student examination data into a clean, structured, and analysis-ready dataset. Power Query was used for data preparation, while DAX was used to create important academic performance measures.

The prepared dataset contains information about students' **gender, race/ethnicity, parental level of education, lunch type, test preparation course, and scores in Mathematics, Reading, and Writing**.

Objectives

- To **import and inspect** the Student Performance dataset in Power BI.
- To **clean the dataset** by handling errors, missing values, duplicates, and inconsistent data.
- To **standardize categorical data** such as Gender, Race/Ethnicity, Parental Level of Education, Lunch, and Test Preparation Course.
- To create calculated columns such as **Total Marks, Average, Performance Grade, and Pass/Fail**.
- To **create a unique Student ID** and improve data consistency using **Capitalize, Trim, and Clean** transformations.
- To **unpivot Math, Reading, and Writing scores** into a structured Student Marks table with **Subject** and **Score** columns.
- To remove **unnecessary or duplicate columns** and organize the data model.
- To create basic **DAX measures** for academic performance analysis.

Data Import and Initial Inspection

The **Student Performance dataset** was imported into Power BI and opened in **Power Query Editor** for initial inspection. Important categorical fields such as **Gender, Race/Ethnicity, Parental Level of Education, Lunch, and Test Preparation** were reviewed. The **Column Quality** indicators showed **100% valid values, 0% errors, and 0% empty values**, confirming that the data was complete. The initial steps included **Promoted Headers** and **Changed Type**, ensuring the categorical fields were correctly recognized as **Text** and prepared for further data cleaning and transformation.

= Table.TransformColumnTypes("#Promoted Headers",{{"gender", type text}, {"race/ethnicity", type text}, {"parental lev						
	AB gender	AB race/ethnicity	AB parental level of education	AB lunch	AB test preparation course	
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	female	group B	bachelor's degree	standard	none	
2	female	group C	some college	standard	completed	
3	female	group B	master's degree	standard	none	
4	male	group A	associate's degree	free/reduced	none	
5	male	group C	some college	standard	none	
6	female	group B	associate's degree	standard	none	
7	female	group B	some college	standard	completed	
8	male	group B	some college	free/reduced	none	
9	male	group D	high school	free/reduced	completed	
10	female	group B	high school	free/reduced	none	
11	male	group C	associate's degree	standard	none	
12	male	group D	associate's degree	standard	none	
13	female	group B	high school	standard	none	
14	male	group A	some college	standard	completed	
15	female	group A	master's degree	standard	none	
16	female	group C	some high school	standard	none	
17	male	group C	high school	standard	none	
18	female	group B	some high school	free/reduced	none	

Creating Student ID

A **Student ID** column was created to provide a **unique identification number for each student record**. This helps in identifying individual students easily and makes the dataset more organized for further analysis and visualization in Power BI.

Capitalizing Each Word

The **Capitalize Each Word** transformation was applied to improve the **formatting and readability of text values**. It changed values into a proper format, such as

female to **Female**, group b to **Group B**, and bachelor's degree to **Bachelor's Degree**. This also helps maintain consistency across categorical data.

Trimming Text

The **Trim** transformation was used to **remove unnecessary spaces** from the beginning and end of text values. This ensures that categories are stored consistently and prevents extra spaces from creating duplicate or incorrect categories during analysis.

Cleaning Text

The **Clean** transformation was applied to **remove unwanted or non-printable characters** from text fields. This improves the quality and consistency of the dataset and ensures that the categorical values are clean, readable, and suitable for further Power BI analysis.

= Table.RenameColumns("#Added Conditional Column1",{{"gender", "Gender"}, {"race/ethnicity", "Race/Ethnicity"}, {"parental level of education", "Parental Level of Education"}, {"lunch", "Lunch"}, {"test preparation course", "Test Preparation Course"}})									
123 Student ID	AB Gender	AB Race/Ethnicity	AB Parental Level of Education	AB Lunch	AB Test Preparation Course				
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%				
1	1 Female	Group B	Bachelor'S Degree	Standard	NONE				
2	2 Female	Group C	Some College	Standard	COMPLETED				
3	3 Female	Group B	Master'S Degree	Standard	NONE				
4	4 Male	Group A	Associate'S Degree	Free/Reduced	NONE				
5	5 Male	Group C	Some College	Standard	NONE				
6	6 Female	Group B	Associate'S Degree	Standard	NONE				
7	7 Female	Group B	Some College	Standard	COMPLETED				
8	8 Male	Group B	Some College	Free/Reduced	NONE				
9	9 Male	Group D	High School	Free/Reduced	COMPLETED				
10	10 Female	Group B	High School	Free/Reduced	NONE				
11	11 Male	Group C	Associate'S Degree	Standard	NONE				
12	12 Male	Group D	Associate'S Degree	Standard	NONE				
13	13 Female	Group B	High School	Standard	NONE				
14	14 Male	Group A	Some College	Standard	COMPLETED				
15	15 Female	Group A	Master'S Degree	Standard	NONE				
16	16 Female	Group C	Some High School	Standard	NONE				
17	17 Male	Group C	High School	Standard	NONE				
18	18 Female	Group B	Some High School	Free/Reduced	NONE				

Creating the Total Marks Column

A **Custom Column** named **Total Marks** was created in Power Query to calculate the total marks obtained by each student. The formula **[math score] + [reading score] + [writing score]** combines the student's Math, Reading, and Writing scores to calculate their overall marks. Power Query showed **"No syntax errors have been detected,"** confirming that the formula was entered correctly. The Total Marks column provides a combined measure of student performance and can be

used for calculating the **Average Score**, **Performance Grade**, and other academic performance indicators.

Custom Column

Add a column that is computed from the other columns.

New column name
Total Marks

Custom column formula ⓘ
= [math score] + [reading score] + [writing score]

Available columns
Student ID
gender
race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Creating the Average Score Column

A **Custom Column** named **Average** was created in Power Query using the formula **[Total Marks] / 3**. This calculates the average score of each student across Mathematics, Reading, and Writing. The formula was validated successfully with no syntax errors. The Average column provides a single measure of overall student performance and can be used for further analysis, grading, and dashboard visualization.

Custom Column

Add a column that is computed from the other columns.

New column name
Average

Custom column formula ⓘ
= [Total Marks]/3

Available columns
Student ID
gender
race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Creating the Performance Grade Column

A **Conditional Column** named **Performance Grade** was created in Power Query based on the student's **Average** score. Students with an average score **greater than or equal to 90** are classified as **Grade A**, scores from **75 to below 90** as **Grade B**, scores from **50 to below 75** as **Grade C**, scores from **40 to below 50** as **Grade D**, and scores **below 40** are classified as **Fail**. This classification helps group students according to their academic performance and makes it easier to analyze grade distribution in the dashboard.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Performance Grade

	Column Name	Operator	Value		Output
If	Average	is greater than or...	90	Then	Grade A
Else If	Average	is greater than or...	75	Then	Grade B
Else If	Average	is greater than or...	50	Then	Grade C
Else If	Average	is greater than or...	40	Then	Grade D
Else If	Average	is less than	40	Then	Fail

Add Clause

Else
null

OK Cancel

Creating the Pass / Fail Column

A **Conditional Column** named **Pass / Fail** was created in Power Query based on the student's **Average** score. Students with an average score **greater than 40** are classified as **Pass**, while students with an average score of **40 or below** are classified as **Fail**. This classification helps identify students who meet the defined passing criterion and supports further analysis of **pass rates and academic performance** in the dashboard.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Pass / Fail

	Column Name	Operator	Value		Output
If	Average	is greater than	40	Then	Pass

Add Clause

Else
Fail

OK Cancel

Creating the Student Marks Table

In the **Student Marks** table, the **Unpivot Columns** transformation was used to restructure the subject-wise marks for easier analysis. The **Math Score, Reading Score, and Writing Score** columns were converted into two columns: **Subject** and **Score**. As a result, each **Student ID** appears three times, with one record for each subject and its corresponding score. The **Subject** column was also formatted consistently as **Math Score, Reading Score, and Writing Score**. This structure makes the **Student Marks** table suitable for subject-wise comparison, performance analysis, and creating interactive Power BI visualizations.

123 Student ID	A6c Subject	123 Score
1	Math Score	72
2	Reading Score	72
3	Writing Score	74
4	Math Score	69
5	Reading Score	90
6	Writing Score	88
7	Math Score	90
8	Reading Score	95
9	Writing Score	93
10	Math Score	47
11	Reading Score	57
12	Writing Score	44
13	Math Score	76
14	Reading Score	78
15	Writing Score	75
16	Math Score	71
17	Reading Score	83
18	Writing Score	78

Creating the Student Details Table

In the **Students Details** table, the **Math Score, Reading Score, and Writing Score** columns were removed because the subject-wise scores had already been separated into the **Student Marks** table using the **Unpivot Columns** transformation. Removing these duplicate columns keeps the **Students Details** table clean and avoids unnecessary data duplication. The table now mainly contains student-related information such as **Student ID, Gender, Race/Ethnicity, Parental Level of Education, Lunch, and Test Preparation Course**, making the data model more organized and suitable for further analysis.

Queries [4] ✕ ✓ fx = Table.RemoveColumns(#Renamed Columns2,{"Math Score", "Reading Score", "Writing Score"})

123 Student ID	APC Gender	APC Race/Ethnicity	APC Parental Level of Education	APC Lunch	APC Test Preparation Course
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	1 Female	Group B	Bachelor'S Degree	Standard	NONE
2	2 Female	Group C	Some College	Standard	COMPLETED
3	3 Female	Group B	Master'S Degree	Standard	NONE
4	4 Male	Group A	Associate'S Degree	Free/Reduced	NONE
5	5 Male	Group C	Some College	Standard	NONE
6	6 Female	Group B	Associate'S Degree	Standard	NONE
7	7 Female	Group B	Some College	Standard	COMPLETED
8	8 Male	Group B	Some College	Free/Reduced	NONE
9	9 Male	Group D	High School	Free/Reduced	COMPLETED
10	10 Female	Group B	High School	Free/Reduced	NONE
11	11 Male	Group C	Associate'S Degree	Standard	NONE
12	12 Male	Group D	Associate'S Degree	Standard	NONE
13	13 Female	Group B	High School	Standard	NONE
14	14 Male	Group A	Some College	Standard	COMPLETED

Preparing the Data Model

After completing the required **data cleaning and transformation**, the resulting tables were reviewed in Power BI to ensure that the data was properly structured for analysis. Unnecessary and redundant columns were removed where appropriate, while the important student and marks information was retained. The transformed data was then loaded into the Power BI data model using **Close & Apply**. After loading, the tables and columns were checked again to confirm that all transformations had been applied correctly and that the dataset was ready for creating **DAX measures, relationships, and dashboard visualizations**.

Creating DAX Measures

1. Average Math Score

The **Average Math Score** measure was created using the AVERAGE function to calculate the mean score obtained by students in Mathematics. This measure helps evaluate the overall performance of students in the Math subject and can be used in KPI cards and dashboard visualizations.

Formula :

Average Math Score = **AVERAGE(StudentsPerformance[Math Score])**

2. Average Reading Score

The **Average Reading Score** measure was created using the AVERAGE function to calculate the mean Reading score of all students. It helps in understanding the

overall performance of students in Reading and allows comparison with other subjects.

Formula :

Average Reading Score = `AVERAGE(StudentsPerformance[Reading Score])`

3. Average Writing Score

The **Average Writing Score** measure calculates the mean score obtained by students in Writing using the `AVERAGE` function. This measure provides an overall view of Writing performance and can be used for subject-wise comparison in the dashboard.

Formula :

Average Writing Score = `AVERAGE(StudentsPerformance[Writing Score])`

4. Overall Average Score

The **Overall Average Score** measure calculates the average of the previously created **Average Score** column. It provides a single indicator of students' overall academic performance across the three subjects and is useful for displaying the overall performance as a KPI.

Formula :

Overall Average Score =

`DIVIDE(
 [Average Math Score] +
 [Average Reading Score] +
 [Average Writing Score],
 3
)`

5. Overall Pass %

The **Overall Pass %** measure calculates the percentage of students classified as **Pass** compared with the total number of students. The `DIVIDE` and `CALCULATE` functions are used to perform the calculation safely. The resulting decimal value,

such as **0.97**, is formatted as a percentage with zero decimal places in Power BI, displaying it as **97%**. This measure is useful for monitoring the overall pass rate in the Student Performance Dashboard.

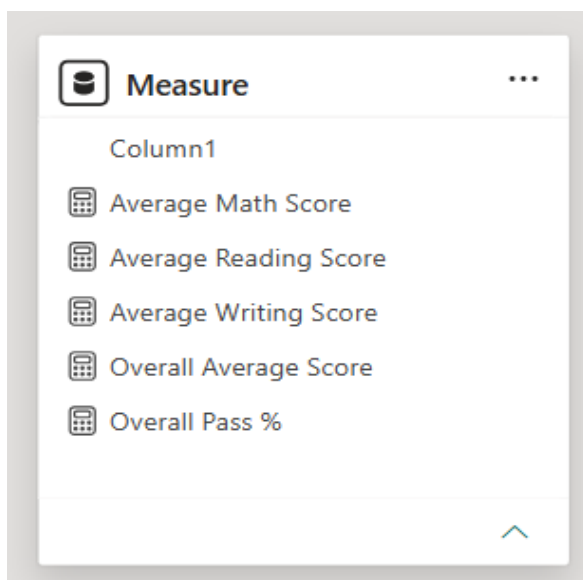
Formula :

Overall Pass % =

```
DIVIDE(  
    CALCULATE(  
        COUNTROWS('StudentsPerformance'),  
        'StudentsPerformance'[Pass / Fail] = "Pass"  
    ),  
    COUNTROWS('StudentsPerformance'),  
    0  
)
```

Measure Table

A separate **Measure** table was created in Power BI to organize and store the main DAX measures used for the Student Performance Dashboard. The table contains **Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score, and Overall Pass %**. Keeping these measures in a dedicated table makes the data model more organized and makes the measures easier to locate and use while creating dashboard visuals and KPI cards.

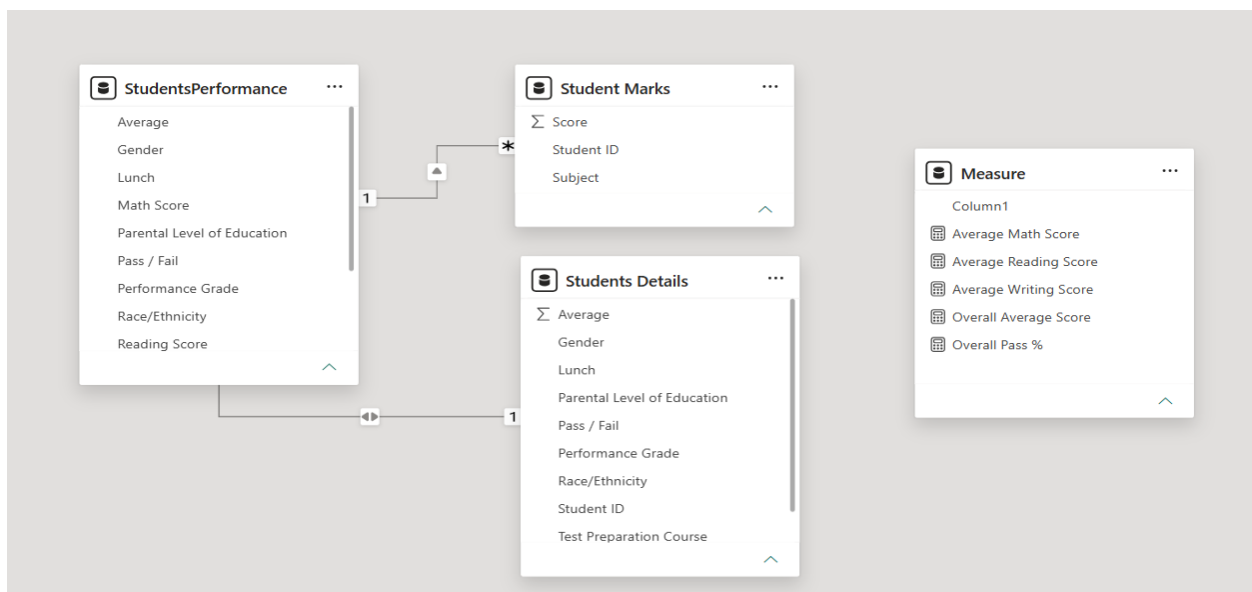


Final Data Model Relationships

The final Power BI data model consists of **StudentsPerformance**, **Student Marks**, **Students Details**, and **Measure** tables. The **StudentsPerformance** table is connected to the other student-related tables using **Student ID**.

A **one-to-many (1:*) relationship** connects **StudentsPerformance** with **Student Marks**, linking student information with subject-wise scores. A **one-to-one (1:1) relationship** connects **StudentsPerformance** with **Students Details**, linking performance data with details such as gender, lunch, parental education, race/ethnicity, pass/fail, and test preparation course.

The **Measure** table contains calculated measures such as **Average Math Score**, **Average Reading Score**, **Average Writing Score**, **Overall Average Score**, and **Overall Pass %**. These relationships provide a structured data model for analysing student performance in the Power BI dashboard.



Conclusion

Week 7 successfully prepared the **Student Performance dataset** for analysis in Power BI. Data cleaning and transformation improved the consistency of the data, while calculated columns, unpivoting, and DAX measures provided useful academic performance information. The final data model established clear relationships between **student details**, **subject marks**, and **performance data**, creating a structured and reliable foundation for student performance analysis.